

THE ADVANTAGES AND DISADVANTAGES OF TECHNOLOGY

Is technology making our lives better — or more complicated? • LEVEL A



Technology shapes almost every part of modern life — but does it always make things better?

The second half of the 20th century saw the start of a new **revolution** in technology that seemed just as powerful as the industrial revolution that **preceded** it. These changes are still ongoing and are not easy to assess — in many ways, they may only just be beginning. The new technologies that have caused such **upheaval** are **interrelated** and overlap in many ways.

What Are the New Technologies?

- **Computers.** They began to gain a foothold in the 1980s and are now found in almost every part of modern life.
- **New communication technologies.** The internet, mobile phones, and wifi mean that people and information can be accessed from **virtually** anywhere.
- **Digital formats.** The digital format for **storing information** has **revolutionized** how photos, music, videos and documents are stored and shared.
- **Multifunctional devices.** One mobile phone can now be a camera, internet browser, music player, and video player — all in one.
- **New military technology.** Pilotless drones and cyberwarfare are two **controversial** examples of how technology is increasingly used by the military.

1 You need to contact a friend who lives far away. What do you do?

a I write a letter and post it — it's more personal.

b I send a message online or make a video call instantly.

2 You want to learn something new. Where do you go first?

a I go to the library and find a book.

b I search online — I can find almost anything on the internet.

These situations show how technology has changed everyday life. Most people today would choose option b — and that tells us a lot about how **dependent** we have become on modern technology. But is this always a good thing?

■ ADVANTAGES

- Internet and mobile technology let us communicate with anyone in the world.
- **Automation** means less **repetitive** work and greater **efficiency** in many jobs.
- The internet makes information more **accessible** and learning more **affordable**.
- The entertainment industry has been **democratized** — artists no longer need big companies.
- Technology saves time with **automated** banking, washing machines, and smart devices.
- Machines now use less power and function more efficiently than before.

■ DISADVANTAGES

- **Social networking** is replacing face-to-face contact, leading to **cyber-bullying** and alienation.
- **Automation** and **outsourcing** are reducing jobs and lowering wages for many workers.
- Privacy is harder to protect — financial data can be hacked and identities stolen.
- **Intellectual property** theft makes it hard for creative people to earn money.
- Modern society has become deeply **dependent** on technology and vulnerable to cyber attacks.
- Technology produces **toxic** and non-**biodegradable** waste, harming the environment.

7,000+

languages spoken today yet tech connects us all

20M

jobs estimated to be replaced by robots by 2030

24/7

entertainment now available on demand to anyone

The Advantages of Technology

Thanks to the internet and mobile technologies, we can communicate with anyone in the world and form relationships with people who live far away, rather than being limited to people within our **physical vicinity**. This has great benefits for both home and work — employees can now communicate with colleagues from anywhere, and many people are able to work from home.

The increased **automation** of manufacturing and many other jobs means greater **efficiency**, fewer people doing boring, **repetitive** tasks, and more flexibility around working hours. This freedom also cuts down on **carbon emissions**, as fewer people need to commute.

The wealth of information now available online is enormous. Having access to the internet means having access to almost all of the world's knowledge. This makes learning more **accessible** and **affordable**, helping to create a more educated society. The entertainment industry has also been **democratized** — artists no longer need big companies to fund them, and **subscription** services make entertainment available to far more people.

"The number one benefit of information technology is that it empowers people to do what they want to do. It lets people be creative, productive, and learn things they didn't think they could learn before."

— Steve Ballmer, former CEO of Microsoft

The Disadvantages of Technology

Human relations are **diminished** in the virtual world. As **social networking** increasingly replaces real face-to-face contact, problems such as **cyber-bullying**, online stalking, and cybercrime are growing — all made worse by the anonymity of the internet. A lack of physical contact can also lead to feelings of loneliness, depression, and anxiety.

Human beings are becoming less valued in the workplace as their roles are replaced by machines. **Outsourcing** has brought bigger profits for companies, but fewer jobs and lower wages for ordinary workers. It is estimated that 20 million jobs will be replaced by robots by 2030, which means we must use our **innovativeness** to create new opportunities.

Privacy is increasingly difficult to protect. Financial details can be hacked, personal photos posted without permission, and identities stolen. **Intellectual property** theft and piracy make it hard for creative people to earn money from their work. Society has also grown deeply **dependent** on technology — hospitals, airports, and power grids could all be knocked out by a cyber attack. Meanwhile, many devices are made from **toxic** or non-**biodegradable** materials, creating serious environmental problems.

"Technology is supposed to make our lives easier, allowing us to do things more quickly and efficiently. But too often it seems to make things harder."

— James Surowiecki, American journalist

Technology: A Double-Edged Sword

The technological **revolution** will only continue — and we must learn how to adapt. Every advancement brings both advantages and disadvantages, so it is impossible to say that technology is simply good or bad. The only thing we can do is try to use our new knowledge and tools to improve our world, rather than harm it.

ACTIVITIES

1

Read the article and answer True (T) or False (F).

- 1 The technological revolution is less important than the industrial revolution.
- 2 Mobile phones can now perform many tasks that used to need separate devices.
- 3 Automation means workers always have to do more repetitive jobs.
- 4 The internet makes it possible to access almost all the world's knowledge.
- 5 It is estimated that 20 million jobs will be replaced by robots by 2030.
- 6 Subscription services have made entertainment less available to ordinary people.

2

Match the highlighted words in the article to these meanings.

- 1 came before something; happened earlier
- 2 major, sudden change that causes disorder
- 3 connected to each other in several ways
- 4 made available to everyone, not just a few
- 5 the process of machines doing jobs instead of people
- 6 sending company work to another country to save money

3

Read the article again and answer these questions.

- 1 Give two examples of how mobile technology has improved work life.
- 2 How has the internet changed the entertainment industry for artists?
- 3 What does the article say could happen to society if technology were taken away?
- 4 Why does the article call technology 'a double-edged sword'?

4

Think about your own life. Write 4–6 sentences about how technology affects you.

Use these phrases to help you:

Technology helps me ... / One problem with technology is ... / I think in the future ...

■ TALKING POINTS

Do you think technology makes your life easier or more stressful? Give examples.

What would your day look like without any technology at all?

Do you think robots taking over jobs is a good or bad thing? Why?

How can we use technology to help the environment instead of harming it?